1. Bézout's identity:

$$\gcd(a,b)=d \implies \exists x,y\in\mathbb{Z}: ax+by=d. \tag{1}$$

From (1) and $\gcd(a,b)=1$ we have:

$$ax+by=1$$

By multiplying right part of $a|c$ by the equation above we get:

$$a \mid c \iff a \mid c(ax+by) \iff a \mid acx+bcy$$

$a \mid acx$ is obviously true.
From the statement: $a \mid bc \implies a \mid bcx$
Therefore $a \mid c$

2. Let's assume:

$$S=\{p:\text{prime},\ p\equiv 3 \pmod 4\}=\{p_1,\dots,p_N\}$$

for some finite N.
Let's consider the following product:

$$P=\prod_{i=1}^{N} p_i$$

Let's consider two cases:

(a)
$$P\equiv 1 \pmod 4$$

In this case we can take $P'=P+2$.

$$\forall i: 1\le i\le N: \quad P'\equiv 2 \pmod{p_i}$$
$$P'\equiv 3 \pmod 4$$

(b)

$$P \equiv 3 \mod 4$$

In this case let's consider $P' = P * 3 + 2$. P' satisfies same criteria as in previous case.

Now let's consider arbitrary number $n = q_1 \dots q_m,$ where q_i are primes . If $n \equiv 3 \pmod 4$, then for its factors the following holds:

(a)

$$\forall i : q_i \neq 2 \pmod 4$$

(b)

$$\exists i : q_i \equiv 3 \pmod 4$$

Since P' is not divisible by any elements of S but it must have a factor that satisfies conditions above it means contradiction, therefore S is not finite.

3.

$$D(n) = \{d \in \mathbb{N} \colon d \mid n\}$$
$$d(n) = \#D(n)$$
$$D(n) \times D(m) = \{(x, y) \colon x \mid n, y \mid m, x \in \mathbb{N}, y \in \mathbb{N}\}$$

Need to prove the following:

$$f \colon D(n) \times D(m) \xrightarrow{\sim} D(nm)$$
$$f(x, y) = xy$$
$$x \mid n, y \mid m \implies n = xr, m = ys \implies nm = xy(rs) \implies xy|nm$$

(a) for injection suppose that it doesn't hold:

$$\exists (x, y), (x', y') \colon (x \neq x' \text{ or } \ y \neq y') \text{ and } xy = x'y'$$

To rule out case $x = x', y \neq y'$ (and symmetric one):

$$\begin{cases} xy = x'y' \\ x = x' \end{cases} \implies y = y'$$

contradiction, therefore $x \neq x'$ and $y \neq y'$.

Then

$$\text{let } P = \{p \in \mathbb{N} : p \text{ is prime}, p \mid nm\} = \{p_1 \dots p_k\}$$
$$x = \prod_{i=1}^{k} p_i^{a_i} \qquad y = \prod_{i=1}^{k} p_i^{b_i} \qquad x' = \prod_{i=1}^{k} p_i^{a_i'} \qquad y' = \prod_{i=1}^{k} p_i^{b_i'}$$
$$\forall i\colon a_i + b_i = a_i' + b_i'$$
$$\exists i\colon a_i \neq a_i'$$
$$\gcd(n, m) = 1 \implies \forall i\colon \begin{cases} a_i = 0 \lor b_i = 0 \\ a_i' = 0 \lor b_i' = 0 \end{cases}$$

therefore

$$\text{either } \forall i : a_i = a_i', b_i = b_i'$$
$$\text{or } \exists i : a_i > 0, b_i = 0, a_i' = 0, b_i' > 0$$
$$\text{or} \exists i : a_i = 0, b_i > 0, a_i' > 0, b_i' = 0$$

but then (assume second case without loss of generality):

$$p_i \mid \gcd(x, y')$$
$$x \mid n, y' \mid m \implies \gcd(x, y') \mid \gcd(n, m)$$
$$p_i \mid \gcd(n, m)$$

contradiction. Therefore f is injection.

(b) for surjection consider arbitrary $z \mid nm$:

$$z = \prod_{i=1}^{k} p_i^{c_i} \qquad n = \prod_{i=1}^{k} p_i^{a_i} \qquad m = \prod_{i=1}^{k} p_i^{b_i}$$

then

$$x = \prod_{i=1}^{k} p_i^{\min(a_i, c_i)} \qquad y = \prod_{i=1}^{k} p_i^{\min(b_i, c_i)}$$
$$x \mid n, y \mid m$$
$$\begin{cases} \gcd(n, m) = 1 \implies \forall i : a_i = 0 \text{ or } b_i = 0 \\ z|nm \implies \forall i : c_i \leq a_i + b_i \end{cases} \implies \text{xy=z}$$

therefore for every arbitrary z exists (x, y) such that $f(x, y) = z$, f is surjection.

Since f is both injection and surjection it's a bijection.